

QUESTIONS

(There are 3 questions in this exam, you should submit your document(s) until 14:15)

1. **(30 points)** Consider the below relational schema where a book can be written by one or more authors, and a book is published by only one publisher. A publisher can publish any number of books. The underlined attributes are the primary keys of the tables.

BOOKS (BookID, Title, PublisherID, Year, Price)

AUTHORS (AuthorID, Name, WritingStyle)

WRITE (BookID, AuthorID)

PUBLISHERS (PublisherID, Name, Address, Country)

Write the following queries in **relational algebra**.

- a) Find the titles and prices of books that are written by author whose name is “XYZ” and published by the publishers in England.
- b) Find the PublisherID, Name, and Address of the publishers that published all books.
- c) Find the names of authors who have not written any book.

2. **(50 points)** Consider the relational schema given in the first question. Write the following queries in **standard SQL**.

- a) Find the titles and prices of books that are written by author whose name is “XYZ” and published by the publishers in England.
- b) Find the titles of books that are written by all authors.
- c) For each publisher, display PublisherID, publisher name, number of books published by the publisher, and average price of the books for the publisher.
- d) Find the title and price of the most expensive book.
- e) Find the titles of books that are written by authors having “horror” or “science-fiction” as writing style.

3. **(20 points)** Consider a relation R with 5 attributes ABCDE. Let the following functional dependencies be given:

$B \rightarrow CD$

$C \rightarrow E$

$DE \rightarrow A$

- a) Which one(s) of A, B, C, and DE can be a (primary) key for relation R? Explain your answer and suggest a primary key for this table.
- b) Is R in BCNF? Explain your answer.

Good Luck...

Duration:60 minutes